Permutations and Combinations :

→ Permutation : "A permutation is an arrangement of a number of objects in some definite order taken some or all at a time." It is denoted by ^nP_r, if n distinct objects and out of them r taken at a time.

Thm-1 The no. of different permutations of n distinct objects taken r at a time, $r \le n$ is given by

$$^nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\ldots\ldots(n-(r-1)).$$

Thm-2 : $^nP_n = n!$, i.e. no. of permutations of n things taken all at a time is $n!$

Ex-1 : Find the value of n if,

(i) $4 \times {}^nP_3 = {}^{(n+1)}P_3$

(ii) $6 \times {}^nP_3 = 3 \times {}^{(n+1)}P_3$

(iii) $3 \times {}^nP_4 = 7 \times {}^{(n-1)}P_4$

[A : (i) 3 (ii) 5 (iii) 7]

Ex-2 : How many variable names of 8 letters can be formed from the letters a, b, c, d, e, f, g, h, i if no letter is repeated. [A → $^9P_8$]

Ex-3 : There are 10 persons called for an interview. 4 posts Each one is capable to be selected for the job. How many permut. are there to select [A → $^{10}P_4$]

Ex-4 How many 6-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with 30. [A → $^7P_4$]

Ex-5 In how many ways 5 different Mathematics books and 4 different Physics books be arranged in a shelf so that all four Physics books are together ? [A → 6! × 4! = 17280]

Ex-6 : How many permutations can be made out of the letter of word " COMPUTER " ? How many of these (i) begin with C ? (ii) end with R ? (iii) Begin with C and end with R ? (iv) C and R occupy the end places ? [A → 40320, 5040, 5040, 720, 1440]

→ Permutations when all the objects are not distinct :-

Thm-3 The no. of permutations of n objects, of which n_1 objects are of one kind and n_2 objects are of another kind, when all are taken at a time is

$$\frac{n!}{n_1! \times n_2!}$$

Ex-1 : Determine the no. of permutations that can be made out of the letters of the word 'PROGRAMMING'.
[A → 4989600]

Ex-2 :- There are 4 blue, 3 red and 2 black pens in a box. These are drawn one by one. Determine all the different permutations. [A → 1080] [1260]

Ex-3 : How many different variable names can be formed by using the letters a, a, a, b, b, b, b, c, c, c ? [A → 4200]

Ex-4 : How many 7-digit numbers can be formed using digits 1, 7, 2, 7, 6, 7, 6 ?
[A → 420]

Permutations with Repeated Objects

Thm-1 :- The no. of different permutations of n distinct objects taken r at a time when every object is allowed to repeat any number of time is given by n^r.

Ex-1: How many 4 digit numbers can be formed by using the digits 2, 4, 6, 8 when repeatation of digits is allowed.
[A → 256]

Ex-2 If repeatation of digits is allowed, how many 2-digits even nos. can be formed by using the digits 1, 3, 4, 6, 8?
[A → 15]

Ex-3 In how many ways can 5 software Projects be allotted to 6 final Year students when all the 5 projects are not allotted to the same student?
[A → 7776]

CIRCULAR PERMUTATIONS

→ Circular permutations are the permutations of the objects placed in a circle.

Thm-1: The number of circular permutations of n different object is $(n-1)!$

Ex-1: In how many ways can the letters a, b, c, d, e, f be arranged in a circle?
(A = 120)

Ex-2 In how many ways 10 programmers can sit on a round table to discuss the project so that project leader and a particular programmer always sit together?
(A = 80640)

Ex-3: Determine the number of ways in which 5 software engineers and 6 electronics engineers can be sitted at a round table so that no two software eng. can sit together?
(A = 86400)

Combination

A Combination is a selection of some or all, objects from a set of given objects, where order of the objects does not matter." The no. of combinations of n objects, taken r at a time is represented by ^nC_r or $C(n, r)$.

Thm-1: The number of Combinations of n different things, taken r at a time is given by

$$^nC_r = \frac{n!}{r!\,(n-r)!}, \quad n \geq r \geq 1$$

Rmk-1 : $^nC_n = 1$

Rmk-2 : $^nC_0 = 1$

Rmk-3 : $^nC_r = {}^nC_{n-r}$, $n \geq r \geq 1$

Ex-1: Find the value of n if,
(i) $^nC_{n-2} = 10$ & (ii) $^{20}C_{n+2} = {}^{20}C_{2n-1}$

(A → (i) 5 , (ii) 3, 8)

Ex-2: How many 16 - bit strings are there containing exactly five 0's?
(A - 4368)

Ex-3: From 10 programmers in how many ways can 5 be selected when (i) A particular programmer is included every time & (ii) A part. prog. is not included at all. [A → 126 , 126]

Combination

Ex-4 : show that $^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$, where $n \geq r \geq 1$ and n and r are natural numbers